

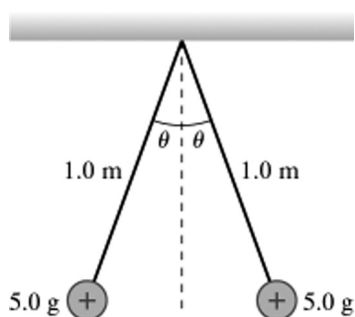
area of cylinder: $2\pi R L$ volume of cylinder : $\pi R^2 L$ area sphere: $4\pi R^2$ micro is 10^{-6} 1g=0.001 kg
 volume sphere $\frac{4}{3}\pi R^3$ 1m = 1000mm 1m = 100cm pico(p) is 10^{-12} nano(n) is 10^{-9}

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) Two charges that are separated by one meter exert 1 - N forces on each other. If the charges are pushed together so the separation is 25 centimeters, the force on each charge will be 1) _____
 A) 16 N. B) 1 N. C) 2 N. D) 8 N. E) 4 N.

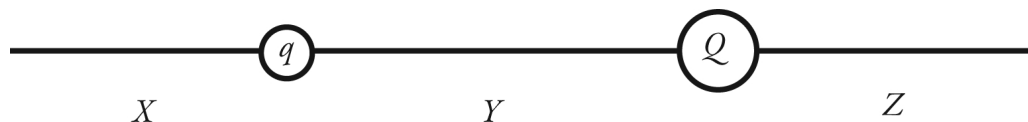
- 2) A $+5.0\text{-}\mu\text{C}$ point charge is placed at the 0 cm mark of a meter stick and a $-4.0\text{-}\mu\text{C}$ charge is placed at the 50 cm mark. What is the net electric field at the 30 cm mark? ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$) 2) _____
 A) $1.4 \times 10^6 \text{ N/C}$ B) $5.0 \times 10^5 \text{ N/C}$
 C) $4.0 \times 10^5 \text{ N/C}$ D) $9.0 \times 10^5 \text{ N/C}$

- 3) The figure shows two tiny 5.0-g spheres suspended from very light 1.0-m-long threads. The spheres repel each other after each one is given the same positive charge and hang at rest when $\theta = 4.1^\circ$. What is the charge on each sphere? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$) 3) _____



- A) 180 nC B) 89 nC C) 360 nC D) 22 nC E) 45 nC

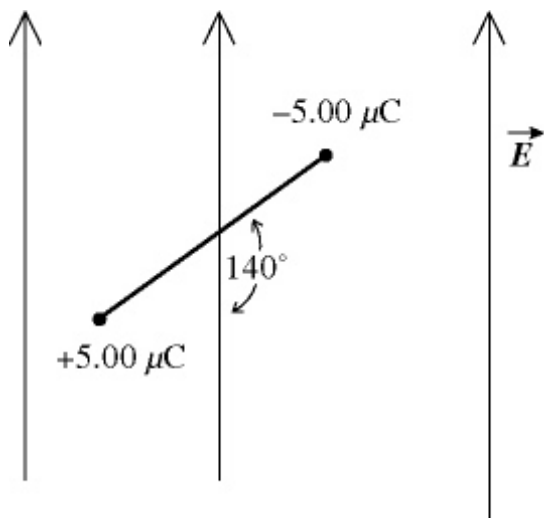
- 4) The figure shows two unequal point charges, q and Q , of opposite sign. Charge Q has greater magnitude than charge q . In which of the regions X, Y, Z will there be a point at which the net electric field due to these two charges is zero? 4) _____



- A) only regions X and Z
 B) only region Y
 C) only region X
 D) only region Z
 E) all three regions

- 5) An electric dipole consists of charges $\pm 5.00 \mu\text{C}$ separated by 1.20 mm . It is placed in a vertical electric field of magnitude 525 N/C oriented as shown in the figure. The magnitude of the net torque this field exerts on the dipole is closest to

5) _____



- A) $1.01 \times 10^{-6} \text{ N} \cdot \text{m}$.
- B) $2.02 \times 10^{-6} \text{ N} \cdot \text{m}$.
- C) $2.41 \times 10^{-6} \text{ N} \cdot \text{m}$.
- D) $3.15 \times 10^{-6} \text{ N} \cdot \text{m}$.
- E) $1.21 \times 10^{-6} \text{ N} \cdot \text{m}$.

- 6) Building up on previous question. The dipole

- A) does not twist
- B) vibrates
- C) rotates CCW
- D) rotates CW

6) _____

- 7) If the electric field just outside a thin conducting sheet is equal to 1.5 N/C , determine the surface charge density on the conductor.

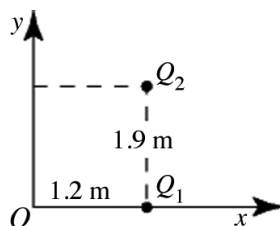
7) _____

- A) 35 pC/m^2
- B) 27 pC/m^2
- C) 13 pC/m^2
- D) 53 pC/m^2

- 8) Two point charges, $Q_1 = -1.0 \mu\text{C}$ and $Q_2 = +4.0 \mu\text{C}$, are placed as shown in the figure.

8) _____

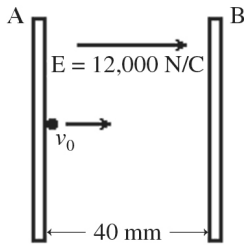
($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$) The y component of the electric field, at the origin O , is closest to



- A) $6.0 \times 10^3 \text{ N/C}$.
- B) $3.8 \times 10^3 \text{ N/C}$.
- C) $-3.8 \times 10^3 \text{ N/C}$.
- D) $7.1 \times 10^3 \text{ N/C}$.
- E) $-6.0 \times 10^3 \text{ N/C}$.

- 9) A proton is placed in an electric field of intensity 700 N/C. What are the magnitude and direction of the acceleration of this proton due to this field? ($m_{\text{proton}} = 1.67 \times 10^{-27}$ kg, $e = 1.60 \times 10^{-19}$ C) 9) _____
- A) 6.71×10^9 m/s² opposite to the electric field
 B) 67.1×10^{10} m/s² in the direction of the electric field
 C) 6.71×10^{10} m/s² in the direction of the electric field
 D) 6.71×10^{10} m/s² opposite to the electric field
 E) 67.1×10^{10} m/s² opposite to the electric field
- 10) Two large, flat, horizontally oriented plates are parallel to each other, a distance d apart. Half way between the two plates the electric field has magnitude E . If the separation of the plates is reduced to $d/2$ what is the magnitude of the electric field half way between the plates? 10) _____
- A) E B) 0 C) $E/2$ D) $4E$ E) $2E$
- 11) A piece of plastic has a net charge of $+2.00 \mu\text{C}$. How many more protons than electrons does this piece of plastic have? ($e = 1.60 \times 10^{-19}$ C) .hint: charge is quantized. easy 11) _____
- A) 1.25×10^{13}
 B) 3.01×10^{23}
 C) 2.50×10^{19}
 D) 1.25×10^{19}
 E) 2.50×10^{13}
- 12) A metal sphere of radius 10 cm carries an excess charge of $+2.0 \mu\text{C}$. What is the magnitude of the electric field 5.0 cm above the sphere's surface? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9$ N · m²/C²) tricky 12) _____
- A) 8.0×10^5 N/C
 B) 8.0×10^7 N/C
 C) 4.0×10^9 N/C
 D) 4.0×10^5 N/C
 E) 4.0×10^7 N/C
- 13) Two 10¢ coins (dimes) carrying identical charges are lying 2.5 m apart on a table. If each of these coins experiences an electrostatic force of magnitude 2.0 N due to the other coin, how large is the charge on each coin? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9$ N · m²/C²). hint: use coulombs law. 13) _____
- A) $6.7 \mu\text{C}$ B) $37 \mu\text{C}$ C) $56 \mu\text{C}$ D) $5.2 \mu\text{C}$ E) $26 \mu\text{C}$

- 14) A pair of charged conducting plates produces a uniform field of 12,000 N/C, directed to the right, between the plates. The separation of the plates is 40 mm. An electron is projected from plate A, directly toward plate B, with an initial speed of $v_0 = 2.0 \times 10^7$ m/s. What is the speed of the electron as it strikes plate B? ($e = 1.6 \times 10^{-19}$ C, $m_{\text{electron}} = 9.11 \times 10^{-31}$ kg)
hint: an electron wants to go against the electric field. So its acceleration is negative.



- A) 2.1×10^7 m/s
B) 1.2×10^7 m/s
C) 1.8×10^7 m/s
D) 2.4×10^7 m/s
E) 1.5×10^7 m/s
- 15) Under electrostatic conditions, the electric field just outside the surface of any charged conductor
A) is always zero because the electric field is zero inside conductors.
B) is always parallel to the surface.
C) is always perpendicular to the surface of the conductor.
D) is perpendicular to the surface of the conductor only if it is a sphere, a cylinder, or a flat sheet.
E) can have nonzero components perpendicular to and parallel to the surface of the conductor.
- 16) An infinitely long **nonconducting** cylinder of radius $R = 2.00$ cm carries a uniform volume charge density of $18.0 \mu\text{C}/\text{m}^3$. Calculate the electric field at distance $r = 1.00$ cm from the axis of the cylinder. ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$)
A) zero
B) 10.2×10^3 N/C
C) 5.10×10^3 N/C
D) 2.00×10^3 N/C
E) 2.50×10^3 N/C
- 17) A spherical, non-conducting shell of inner radius $r_1 = 10$ cm and outer radius $r_2 = 15$ cm carries a total charge $Q = 15 \mu\text{C}$ distributed uniformly throughout the volume of the shell. What is the magnitude of the electric field at a distance $r = 12$ cm from the center of the shell? ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)
A) 5.75×10^6 N/C B) 5.75×10^3 N/C C) 2.87×10^3 N/C D) 2.87×10^6 N/C
- 18) A 16-nC charge is distributed uniformly along the x axis from $x = 0$ to $x = 4$ m. Which of the following integrals is correct for the magnitude (in N/C) of the electric field at $x = +10$ m on the x axis?

A) $\int_0^{10} \frac{154 dx}{x^2}$

B) $\int_0^4 \frac{36 dx}{x^2}$

C) $\int_0^4 \frac{36 dx}{(10-x)^2}$

D) $\int_0^4 \frac{154 dx}{(10-x)^2}$

- 19) Particle A has twice the charge of nearby particle B. Compared to the force on Particle A, the force on Particle B is _____
 19) _____
 A) four times as much.
 B) twice as much.
 C) the same.
 D) half as much.
 E) none of the above
- 20) Two charges of 15 pC and -40 pC are inside a cube with sides that are of 0.40-m length. Determine the net electric flux through the surface of the cube. _____
 20) _____
 A) $-2.8 \text{ N} \cdot \text{m}^2/\text{C}$ B) $2.8 \text{ N} \cdot \text{m}^2/\text{C}$ C) $-1.1 \text{ N} \cdot \text{m}^2/\text{C}$ D) $+1.1 \text{ N} \cdot \text{m}^2/\text{C}$

Electric Field & Force

$$F = \frac{kq_0q}{r^2}$$

$$E = \frac{F}{q} = \frac{kq_0}{r^2}$$

$$E_{ring} = \frac{kQx}{(x^2+a^2)^{3/2}}$$

$$E_{line} = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$E_{disk} = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{R^2+z^2}} \right]$$

$$E_{sheet} = \frac{\sigma}{2\epsilon_0}$$

$$E_{para. plates} = \frac{\sigma}{\epsilon_0}$$

$$p = qd \text{ (dipole moment)}$$

$$\vec{\tau}_{dipole} = \vec{p} \times \vec{E}$$

$$\tau_{dipole} = pE \sin \phi$$

$$U = -pE \cos \phi$$

$$\Phi_E = EA \cos \phi$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

21)

21) _____

Answer Key

Testname: TEST1_25

- 1) A
- 2) A
- 3) B
- 4) C
- 5) B
- 6) D
- 7) B
- 8) A
- 9) C
- 10) A
- 11) A
- 12) A
- 13) B
- 14) E
- 15) C
- 16) B
- 17) D
- 18) C
- 19) C
- 20) A
- 21) FALSE